EXERCISE 3

Canonical Form of Four Systems with Complex Eigenvalues

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Canonical Form • Q = [u | w] • Four Matrices

STATEMENT

For the following equations find the eigenvalues and eigenvectors of the matrix on the right-hand side, and hence find the coordinate transformation that will put the equations into their standard simple (canonical) form. Show that this transformation has the desired effect.

- (i) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} \mathbf{x}$;
- (ii) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} -2 & 3 \\ -6 & 4 \end{pmatrix} \mathbf{x}$;
- (iii) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} -11 & -2 \\ 13 & -9 \end{pmatrix} \mathbf{x}$;
- (iv) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} 7 & -5 \\ 10 & -3 \end{pmatrix} \mathbf{x}$.

PHASE 1: SETUP

The standard (canonical) form for a complex pair $\rho \pm i\omega$ is $\dot{\mathbf{z}} = J\mathbf{z}$ with $J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$, reached by $\mathbf{z} = Q^{-1}\mathbf{x}$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, $\mathbf{u} + i\mathbf{w}$ the eigenvector of $\rho + i\omega$ (Topic 4 Sec. 7.4). For every matrix $\Delta = (\operatorname{tr} A)^2 - 4 \det A < 0$ (checked in each item), so the eigenvalues are complex conjugate.

PHASE 2: FORMULATION

Steps for each matrix: characteristic equation, eigenvalue $\rho + i\omega$, eigenvector via the first row, $\mathbf{u}$, $\mathbf{w}$, relations (7.4), Q and Q^{-1}, and the explicit product $Q^{-1}AQ = J$.

METHOD OF THE CLASS NOTES [NOTES, P. 10]

The construction of the notes: complex diagonalisation $A = PDP^{-1}$ with $P = (\mathbf{v}_+ \mid \mathbf{v}_-)$, then $Q = (\mathbf{v}_1 \mid \mathbf{v}_2) = PM$ with $M = \begin{pmatrix} 1/2 & -i/2 \\ 1/2 & i/2 \end{pmatrix}$, so that $Q^{-1}AQ = M^{-1}P^{-1}APM = M^{-1}DM = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$. The explicit product $Q^{-1}AQ$ computed for each matrix below checks this canonical form.

PHASE 3: DERIVATION

(i) $A = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} 0 - \lambda & -1 \\ 1 & -1 - \lambda \end{vmatrix} = (0 - \lambda)(-1 - \lambda) - (-1)(1)$$

$$= \lambda^2 - (0 + (-1))\lambda + (0 \cdot (-1) - (-1) \cdot 1) = \lambda^2 - (-1)\lambda + (1) = \lambda^2 + \lambda + 1$$

$\operatorname{tr} A = -1$, $\det A = 1$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 1 - 4 = -3$.

$\Delta < 0$: a complex-conjugate pair. $\lambda = \dfrac{-1 \pm \sqrt{-3}}{2} = \dfrac{-1 \pm \sqrt{3}i}{2}$, so

$$\lambda_\pm = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}\, i \qquad (\rho = -\frac{1}{2},\ \omega = \frac{\sqrt{3}}{2} > 0)$$

Eigenvector of the eigenvalue with positive imaginary part

For $\lambda = \rho + i\omega = -\frac{1}{2} + \frac{\sqrt{3}i}{2}$ the first row of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ reads $(\frac{1}{2} - \frac{\sqrt{3}i}{2})\, v_1 + (-1)\, v_2 = 0$ (the coefficient of v_2 is non-zero because $\Delta < 0$ forces $bc < 0$). Choose $v_1 = -1$; then $v_2 = -(\frac{1}{2} - \frac{\sqrt{3}i}{2}) = -\frac{1}{2} + \frac{\sqrt{3}i}{2}$:

$$\mathbf{v} = \begin{pmatrix} -1 \\ -\frac{1}{2} + \frac{\sqrt{3}i}{2} \end{pmatrix} = \underbrace{\begin{pmatrix} -1 \\ -\frac{1}{2} \end{pmatrix}}_{\mathbf{u}} + i \underbrace{\begin{pmatrix} 0 \\ \frac{\sqrt{3}}{2} \end{pmatrix}}_{\mathbf{w}}$$

Check with the second row: $1 \cdot (-1) + (-\frac{1}{2} - \frac{\sqrt{3}i}{2})(-\frac{1}{2} + \frac{\sqrt{3}i}{2}) = -1 + 1 = 0$ (using $i^2 = -1$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$A\mathbf{u} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = (-\frac{1}{2}) \begin{pmatrix} -1 \\ -\frac{1}{2} \end{pmatrix} - \frac{\sqrt{3}}{2} \begin{pmatrix} 0 \\ \frac{\sqrt{3}}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$$

$$A\mathbf{w} = \begin{pmatrix} -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = \frac{\sqrt{3}}{2} \begin{pmatrix} -1 \\ -\frac{1}{2} \end{pmatrix} + (-\frac{1}{2}) \begin{pmatrix} 0 \\ \frac{\sqrt{3}}{2} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}, \mathbf{w}$,

$$Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} -1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

$\det Q = (-1)(\frac{\sqrt{3}}{2}) - (0)(-\frac{1}{2}) = -\frac{\sqrt{3}}{2}$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-\frac{\sqrt{3}}{2}} \begin{pmatrix} \frac{\sqrt{3}}{2} & 0 \\ \frac{1}{2} & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ -\frac{\sqrt{3}}{3} & \frac{2\sqrt{3}}{3} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

$$\begin{aligned}
(Q^{-1}AQ)_{11} &= (-1)\cdot\frac{1}{2} + 0\cdot(-\frac{1}{2}) = -\frac{1}{2} \\
(Q^{-1}AQ)_{12} &= (-1)\cdot(-\frac{\sqrt{3}}{2}) + 0\cdot(-\frac{\sqrt{3}}{2}) = \frac{\sqrt{3}}{2} \\
(Q^{-1}AQ)_{21} &= (-\frac{\sqrt{3}}{3})\cdot\frac{1}{2} + \frac{2\sqrt{3}}{3}\cdot(-\frac{1}{2}) = -\frac{\sqrt{3}}{2} \\
(Q^{-1}AQ)_{22} &= (-\frac{\sqrt{3}}{3})\cdot(-\frac{\sqrt{3}}{2}) + \frac{2\sqrt{3}}{3}\cdot(-\frac{\sqrt{3}}{2}) = -\frac{1}{2}
\end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = -\frac{1}{2}z_1 + \frac{\sqrt{3}}{2}z_2, \qquad \dot{z}_2 = -\frac{\sqrt{3}}{2}z_1 - \frac{1}{2}z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t}r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s}r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\,ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0\sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{-\frac{1}{2}t}\left(c_1\cos\frac{\sqrt{3}}{2}t + c_2\sin\frac{\sqrt{3}}{2}t\right), \qquad z_2(t) = e^{-\frac{1}{2}t}\left(c_2\cos\frac{\sqrt{3}}{2}t - c_1\sin\frac{\sqrt{3}}{2}t\right)$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

$$\mathbf{x}(t) = e^{-\frac{1}{2}t}\left[\left(c_1\cos\frac{\sqrt{3}}{2}t + c_2\sin\frac{\sqrt{3}}{2}t\right)\begin{pmatrix}-1\\-\frac{1}{2}\end{pmatrix} + \left(c_2\cos\frac{\sqrt{3}}{2}t - c_1\sin\frac{\sqrt{3}}{2}t\right)\begin{pmatrix}0\\\frac{\sqrt{3}}{2}\end{pmatrix}\right]$$

$$x(t) = -c_1e^{-\frac{t}{2}}\cos\frac{\sqrt{3}t}{2} - c_2e^{-\frac{t}{2}}\sin\frac{\sqrt{3}t}{2}, \qquad y(t) = -\frac{\sqrt{3}c_1e^{-\frac{t}{2}}\sin\frac{\sqrt{3}t}{2}}{2} - \frac{c_1e^{-\frac{t}{2}}\cos\frac{\sqrt{3}t}{2}}{2} - \frac{c_2e^{-\frac{t}{2}}\sin\frac{\sqrt{3}t}{2}}{2} + \frac{\sqrt{3}c_2e^{-\frac{t}{2}}\cos\frac{\sqrt{3}t}{2}}{2}$$

The canonical system is $\dot{z}_1 = -\frac{1}{2}z_1 + (\frac{\sqrt{3}}{2})z_2$, $\dot{z}_2 = -(\frac{\sqrt{3}}{2})z_1 + -\frac{1}{2}z_2$: a rotation clockwise with angular speed $\omega = \frac{\sqrt{3}}{2}$ combined with a radial factor $e^{-\frac{1}{2}t}$.

(ii) $A = \begin{pmatrix}-2 & 3\\ -6 & 4\end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix}-2-\lambda & 3\\ -6 & 4-\lambda\end{vmatrix} = (-2-\lambda)(4-\lambda) - (3)(-6)$$

$$= \lambda^2 - (-2+4)\lambda + (-2\cdot 4 - 3\cdot(-6)) = \lambda^2 - (2)\lambda + (10) = \lambda^2 - 2\lambda + 10$$

$\operatorname{tr} A = 2$, $\det A = 10$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4\det A = 4 - 40 = -36$.

$\Delta < 0$: a complex-conjugate pair. $\lambda = \dfrac{2 \pm \sqrt{-36}}{2} = \dfrac{2 \pm 6i}{2}$, so

$$\lambda_{\pm} = 1 \pm 3\,i \qquad (\rho = 1,\ \omega = 3 > 0)$$

Eigenvector of the eigenvalue with positive imaginary part

For $\lambda = \rho + i\omega = 1 + 3i$ the first row of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ reads $(-3 - 3i)\,v_1 + 3\,v_2 = 0$ (the coefficient of v_2 is non-zero because $\Delta < 0$ forces $bc < 0$). Choose $v_1 = 3$; then $v_2 = -(-3 - 3i) = 3 + 3i$:

$$\mathbf{v} = \begin{pmatrix} 3 \\ 3 + 3i \end{pmatrix} = \underbrace{\begin{pmatrix} 3 \\ 3 \end{pmatrix}}_{\mathbf{u}} + i \underbrace{\begin{pmatrix} 0 \\ 3 \end{pmatrix}}_{\mathbf{w}}$$

Check with the second row: $-6 \cdot 3 + (3 - 3i)(3 + 3i) = -18 + 18 = 0$ (using $i^2 = -1$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$A\mathbf{u} = \begin{pmatrix} 3 \\ -6 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 1\begin{pmatrix} 3 \\ 3 \end{pmatrix} - 3\begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -6 \end{pmatrix}$$

$$A\mathbf{w} = \begin{pmatrix} 9 \\ 12 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 3\begin{pmatrix} 3 \\ 3 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 9 \\ 12 \end{pmatrix}$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix}$$

$\det Q = (3)(3) - (0)(3) = 9$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{9}\begin{pmatrix} 3 & 0 \\ -3 & 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & 0 \\ -\frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} 3 & 9 \\ -6 & 12 \end{pmatrix}$$

$$\begin{aligned}
(Q^{-1}AQ)_{11} &= \frac{1}{3}\cdot 3 + 0\cdot(-6) = 1 \\
(Q^{-1}AQ)_{12} &= \frac{1}{3}\cdot 9 + 0\cdot 12 = 3 \\
(Q^{-1}AQ)_{21} &= (-\frac{1}{3})\cdot 3 + \frac{1}{3}\cdot(-6) = -3 \\
(Q^{-1}AQ)_{22} &= (-\frac{1}{3})\cdot 9 + \frac{1}{3}\cdot 12 = 1
\end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 1 & 3 \\ -3 & 1 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = z_1 + 3z_2, \qquad \dot{z}_2 = -3z_1 + z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t}r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s}r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\,ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0\sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{1t}\left(c_1\cos 3t + c_2\sin 3t\right), \qquad z_2(t) = e^{1t}\left(c_2\cos 3t - c_1\sin 3t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

$$\mathbf{x}(t) = e^{1t}\left[\left(c_1\cos 3t + c_2\sin 3t\right)\begin{pmatrix}3\\3\end{pmatrix} + \left(c_2\cos 3t - c_1\sin 3t\right)\begin{pmatrix}0\\3\end{pmatrix}\right]$$

$$x(t) = 3c_1e^{t}\cos 3t + 3c_2e^{t}\sin 3t, \qquad y(t) = -3c_1e^{t}\sin 3t + 3c_1e^{t}\cos 3t + 3c_2e^{t}\sin 3t + 3c_2e^{t}\cos 3t$$

The canonical system is $\dot z_1 = 1z_1 + (3)z_2$, $\dot z_2 = -(3)z_1 + 1z_2$: a rotation clockwise with angular speed $\omega = 3$ combined with a radial factor e^{1t}.

(iii) $A = \begin{pmatrix}-11 & -2\\ 13 & -9\end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A-\lambda I) = \begin{vmatrix}-11-\lambda & -2\\ 13 & -9-\lambda\end{vmatrix} = (-11-\lambda)(-9-\lambda) - (-2)(13)$$

$$= \lambda^2 - (-11 + (-9))\lambda + (-11\cdot(-9) - (-2)\cdot 13) = \lambda^2 - (-20)\lambda + (125) = \lambda^2 + 20\lambda + 125$$

$\operatorname{tr}A = -20$, $\det A = 125$, discriminant $\Delta = (\operatorname{tr}A)^2 - 4\det A = 400 - 500 = -100$.

$\Delta < 0$: a complex-conjugate pair. $\lambda = \dfrac{-20 \pm \sqrt{-100}}{2} = \dfrac{-20 \pm 10i}{2}$, so

$$\lambda_{\pm} = -10 \pm 5\,i \qquad (\rho = -10,\ \omega = 5 > 0)$$

Eigenvector of the eigenvalue with positive imaginary part

For $\lambda = \rho + i\omega = -10 + 5i$ the first row of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ reads $(-1 - 5i)\,v_1 + (-2)\,v_2 = 0$ (the coefficient of v_2 is non-zero because $\Delta < 0$ forces $bc < 0$). Choose $v_1 = -2$; then $v_2 = -(-1 - 5i) = 1 + 5i$:

$$\mathbf{v} = \begin{pmatrix} -2 \\ 1 + 5i \end{pmatrix} = \underbrace{\begin{pmatrix} -2 \\ 1 \end{pmatrix}}_{\mathbf{u}} + i\,\underbrace{\begin{pmatrix} 0 \\ 5 \end{pmatrix}}_{\mathbf{w}}$$

Check with the second row: $13 \cdot (-2) + (1 - 5i)(1 + 5i) = -26 + 26 = 0$ (using $i^2 = -1$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$A\mathbf{u} = \begin{pmatrix} 20 \\ -35 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = (-10)\begin{pmatrix} -2 \\ 1 \end{pmatrix} - 5\begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 20 \\ -35 \end{pmatrix}$$

$$A\mathbf{w} = \begin{pmatrix} -10 \\ -45 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 5\begin{pmatrix} -2 \\ 1 \end{pmatrix} + (-10)\begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} -10 \\ -45 \end{pmatrix}$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} -2 & 0 \\ 1 & 5 \end{pmatrix}$$

$\det Q = (-2)(5) - (0)(1) = -10$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-10}\begin{pmatrix} 5 & 0 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & 0 \\ \frac{1}{10} & \frac{1}{5} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} 20 & -10 \\ -35 & -45 \end{pmatrix}$$

$$\begin{aligned}
(Q^{-1}AQ)_{11} &= (-\frac{1}{2}) \cdot 20 + 0 \cdot (-35) = -10 \\
(Q^{-1}AQ)_{12} &= (-\frac{1}{2}) \cdot (-10) + 0 \cdot (-45) = 5 \\
(Q^{-1}AQ)_{21} &= \frac{1}{10} \cdot 20 + \frac{1}{5} \cdot (-35) = -5 \\
(Q^{-1}AQ)_{22} &= \frac{1}{10} \cdot (-10) + \frac{1}{5} \cdot (-45) = -10
\end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -10 & 5 \\ -5 & -10 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = -10z_1 + 5z_2, \qquad \dot{z}_2 = -5z_1 - 10z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t}r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s}r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\,ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0\sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{-10t}\left(c_1\cos 5t + c_2\sin 5t\right), \qquad z_2(t) = e^{-10t}\left(c_2\cos 5t - c_1\sin 5t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

$$\mathbf{x}(t) = e^{-10t}\left[\left(c_1\cos 5t + c_2\sin 5t\right)\begin{pmatrix}-2\\1\end{pmatrix} + \left(c_2\cos 5t - c_1\sin 5t\right)\begin{pmatrix}0\\5\end{pmatrix}\right]$$

$$x(t) = -2c_1e^{-10t}\cos 5t - 2c_2e^{-10t}\sin 5t, \qquad y(t) = -5c_1e^{-10t}\sin 5t + c_1e^{-10t}\cos 5t + c_2e^{-10t}\sin 5t + 5c_2e^{-10t}\cos 5t$$

The canonical system is $\dot z_1 = -10z_1 + (5)z_2$, $\dot z_2 = -(5)z_1 + -10z_2$: a rotation clockwise with angular speed $\omega = 5$ combined with a radial factor e^{-10t}.

(iv) $A = \begin{pmatrix}7 & -5\\ 10 & -3\end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix}7-\lambda & -5\\ 10 & -3-\lambda\end{vmatrix} = (7-\lambda)(-3-\lambda) - (-5)(10)$$

$$= \lambda^2 - (7 + (-3))\lambda + (7\cdot(-3) - (-5)\cdot 10) = \lambda^2 - (4)\lambda + (29) = \lambda^2 - 4\lambda + 29$$

$\operatorname{tr} A = 4$, $\det A = 29$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4\det A = 16 - 116 = -100$.

$\Delta < 0$: a complex-conjugate pair. $\lambda = \dfrac{4 \pm \sqrt{-100}}{2} = \dfrac{4 \pm 10i}{2}$, so

$$\lambda_\pm = 2 \pm 5\,i \qquad (\rho = 2,\ \omega = 5 > 0)$$

Eigenvector of the eigenvalue with positive imaginary part

For $\lambda = \rho + i\omega = 2 + 5i$ the first row of $(A - \lambda I)\mathbf{v} = \mathbf{0}$ reads $(5 - 5i)\,v_1 + (-5)\,v_2 = 0$ (the coefficient of v_2 is non-zero because $\Delta < 0$ forces $bc < 0$). Choose $v_1 = -5$; then $v_2 = -(5 - 5i) = -5 + 5i$:

$$\mathbf{v} = \begin{pmatrix} -5 \\ -5 + 5i \end{pmatrix} = \underbrace{\begin{pmatrix} -5 \\ -5 \end{pmatrix}}_{\mathbf{u}} + i\,\underbrace{\begin{pmatrix} 0 \\ 5 \end{pmatrix}}_{\mathbf{w}}$$

Check with the second row: $10 \cdot (-5) + (-5 - 5i)(-5 + 5i) = -50 + 50 = 0$ (using $i^2 = -1$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$A\mathbf{u} = \begin{pmatrix} -10 \\ -35 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 2\begin{pmatrix} -5 \\ -5 \end{pmatrix} - 5\begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} -10 \\ -35 \end{pmatrix}$$

$$A\mathbf{w} = \begin{pmatrix} -25 \\ -15 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 5\begin{pmatrix} -5 \\ -5 \end{pmatrix} + 2\begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} -25 \\ -15 \end{pmatrix}$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} -5 & 0 \\ -5 & 5 \end{pmatrix}$$

$\det Q = (-5)(5) - (0)(-5) = -25$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-25}\begin{pmatrix} 5 & 0 \\ 5 & -5 \end{pmatrix} = \begin{pmatrix} -\frac{1}{5} & 0 \\ -\frac{1}{5} & \frac{1}{5} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -10 & -25 \\ -35 & -15 \end{pmatrix}$$

$$\begin{aligned}
(Q^{-1}AQ)_{11} &= (-\frac{1}{5}) \cdot (-10) + 0 \cdot (-35) = 2 \\
(Q^{-1}AQ)_{12} &= (-\frac{1}{5}) \cdot (-25) + 0 \cdot (-15) = 5 \\
(Q^{-1}AQ)_{21} &= (-\frac{1}{5}) \cdot (-10) + \frac{1}{5} \cdot (-35) = -5 \\
(Q^{-1}AQ)_{22} &= (-\frac{1}{5}) \cdot (-25) + \frac{1}{5} \cdot (-15) = 2
\end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 2 & 5 \\ -5 & 2 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = 2z_1 + 5z_2, \qquad \dot{z}_2 = -5z_1 + 2z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t}r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s}r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\,ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0\sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{2t}\left(c_1\cos 5t + c_2\sin 5t\right), \qquad z_2(t) = e^{2t}\left(c_2\cos 5t - c_1\sin 5t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

$$\mathbf{x}(t) = e^{2t}\left[(c_1\cos 5t + c_2\sin 5t)\begin{pmatrix}-5\\-5\end{pmatrix} + (c_2\cos 5t - c_1\sin 5t)\begin{pmatrix}0\\5\end{pmatrix}\right]$$

$$x(t) = -5c_1e^{2t}\cos 5t - 5c_2e^{2t}\sin 5t, \qquad y(t) = -5c_1e^{2t}\sin 5t - 5c_1e^{2t}\cos 5t - 5c_2e^{2t}\sin 5t + 5c_2e^{2t}\cos 5t$$

The canonical system is $\dot z_1 = 2z_1 + (5)z_2$, $\dot z_2 = -(5)z_1 + 2z_2$: a rotation clockwise with angular speed $\omega = 5$ combined with a radial factor e^{2t}.

PHASE 4: INTERPRETATION AND CHECKS

- In each item the explicit product $Q^{-1}AQ$ reproduces J exactly (entry-by-entry arithmetic above).
- (i) has an irrational $\omega = \sqrt{3}/2$ (eigenvalues $-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$, with the surds carried exactly): stable spiral. (ii) $1 \pm 3i$, unstable spiral; (iii) $-10 \pm 5i$, stable spiral (fast decay, ratio $|\rho|/\omega = 2$); (iv) $2 \pm 5i$, unstable spiral.
- The orientation of Q ($\det Q$ sign) decides whether the clockwise rotation of the $\mathbf{z}$-plane stays clockwise in the $\mathbf{x}$-plane; it is $b\omega$ here (positive iff $b > 0$).
- The eigenvalues of each matrix satisfy $\lambda_+ + \lambda_- = \operatorname{tr} A$ and $\lambda_+\lambda_- = \det A$ (checked in the text).